

ERDM — Environmental Reality Distortion Module

Parent Standard: Physical Reality Interpretation Layer (PRIL™)

Category: AI & Interpretation

Subcategory: Environmental Reality Distortion

Type: Physical Interpretation Module

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Status: Canonical · Open Module

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Compatibility: OOF® Methodology OS™ · PRIL™ · CLIA® · RIS · INTEGROS® · SIMULOS®

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Canonical Definition

Environmental Reality Distortion Module defines the methodological conditions under which autonomous systems detect, interpret, and restrict distorted physical-world signals caused by reflections, transparency, glare, refraction, recurring ghost patterns, unstable surfaces, environmental interference, or location-specific false perception conditions before generating physical action.

A system satisfies ERDM only if:

- distorted environmental signals are recognized before execution
- reflective, transparent, or interference-based false interpretation conditions are treated as operational risk
- recurring ghost patterns are identifiable as distortion conditions rather than accepted as valid reality
- environmental distortion uncertainty can restrict, slow, stop, or escalate execution
- physical action is limited when environmental reality interpretation is unreliable or false

A system that treats distorted perception as stable reality does not satisfy ERDM.

Module Function

ERDM defines the distortion-control layer of physical reality interpretation.

It ensures that autonomous systems do not treat every sensed physical signal as real physical truth.

The module applies wherever environmental conditions may create false perception through:

- mirrors
- glass
- reflective surfaces
- glare
- thermal distortion

- weather interference
 - repeated location-specific ghost signals
 - phantom obstacle or phantom motion conditions
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Minimum Implementation Framework (MIF)

Step 1 — Define Distortion-Relevant Conditions

The organization must define which environmental conditions may distort physical interpretation.

Minimum requirement:

- relevant distortion sources are explicit
 - environmental false-signal conditions are identifiable
 - distortion exposure is not treated as undefined background noise
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Step 2 — Define Distortion Recognition Logic

The system must define how distorted perception is recognized.

Minimum requirement:

- reflections, transparency, ghost patterns, or interference are not treated as ordinary signals only
 - distortion recognition conditions are explicit
 - repeated false interpretation is not silently normalized as valid reality
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Step 3 — Define Recurring Ghost Pattern Handling

The system must define how repeated false detections tied to specific locations or environmental patterns are handled.

Minimum requirement:

- recurring ghost conditions are identifiable
- location-bound false interpretation patterns are distinguishable
- repeated false activation is not treated as valid event certainty
- Step 4 — Define Restriction and Escalation Conditions

The system must define when environmental distortion slows, stops, restricts, or escalates execution.

Minimum requirement:

- distortion-related restriction logic is explicit
 - false or unstable environmental interpretation can block physical action
 - continued execution under unresolved distortion is not treated as valid
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Step 5 — Preserve Reality Priority

The system must preserve physical reality integrity above speed, convenience, or optimistic perception assumption.

Minimum requirement:

- distorted perception does not justify continued execution
- convenience does not override distortion uncertainty
- physical action remains subordinate to reliable environmental interpretation

Step 6 — Restrict Invalid Execution

The system must not be treated as valid if physical execution proceeds while environmental distortion remains unresolved, recurring, or structurally ungoverned.

Minimum requirement:

- invalid distortion conditions are identifiable
- unresolved false reality interpretation blocks valid reliance
- sensing presence alone does not restore interpretation validity

Use Case 1 — Reflective Surface and False Obstacle Condition

Scenario

An autonomous vehicle or robot encounters mirrors, reflective structures, glass surfaces, or glare conditions that produce false object, depth, or movement interpretation.

Application

ERDM is used to ensure:

- distortion sources are explicitly recognized
- reflected or transparent false signals are not treated as ordinary reality
- unresolved distortion restricts execution
- physical action is not based on false environmental interpretation

Result

The organization gains a stronger basis for preventing autonomous movement when the system senses an apparent condition that is actually

an environmental distortion.

Use Case 2 — Recurring Ghost Detection at Fixed Location

Scenario

A system repeatedly detects a phantom obstacle, crossing event, or movement warning at the same physical location even though no real object or movement is present.

Application

ERDM is used to ensure:

- recurring ghost patterns are recognized as distortion conditions
 - repeated false activation is not treated as stable evidence of reality
 - execution can be restricted or escalated until interpretation reliability is restored
 - physical behavior remains bounded by governed distortion logic
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Result

The organization gains a structurally governed way to handle persistent environmental false interpretation instead of allowing recurring ghost signals to silently distort execution behavior.

Canonical Closing Statement

**If environmental distortion is interpreted as reality before action,
physical execution is not structurally reliable.**

Related Documents

→ [Physical Reality Interpretation Layer \(PRIL™\)](#)

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Canonical Source

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